SHOBIT VERMA

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EDUCATION

Indian Institute of Technology (IIT), Dhanbad — May 2025
Bachelor of Technology in Mechanical Engineering — Dhanbad, India

EXPERIENCE

SLB (formerly Schlumberger) — Dammam, Saudi Arabia / Pune, India
Mechanical Design Engineer — Jul 2025 – Present

- Liner Hanger Team member responsible for **Mechanical Design, Failure analysis, CAE and sustaining** engineering components
- Completed all the LH trainings for **CAD, DFX, GEMS, SAP, CA, QNs, RANs and pFMEAs** with field training at **SLB barmer** base
- Conducted **BCTC failure** analysis for assemblies, **seals, tolerances and ratings** calculations, disposition of **50+ in-process to field-return assemblies**, Component designs and their reviews, **30+ product issues**, supplier waivers, client SQs and queries.
- Currently assisting **SLB SCTC center, Saudi Arabia** with a recent 2 month visit, working on sustaining LH & MSS product lines.

RMML Lab, National Taiwan University — Taipei, Taiwan
Research Intern under Prof Ping-Lang Yen — Offline: May 2024 – Jul 2024, Later Remote

- Developed a PKM robot for respiratory simulation (Patent ID: 519664-001) from ideation to **design, manufacturing and control**.
- Developed a **6-bar kinematic cutting tool** coupled with a hook-and-pull mechanism for autonomous harvesting (UK ID: 6552819).
- Fabricated and built **341 components** requiring **108 bearings** for the PKM robot. For the Kinematic cutter, developed **20+ prototypes** and finalized design via field test. Derived its governing equation by defining **40 different parameters**.
- Also developed a **RCM-based needle insertion mechanism** for autonomous tumour surgery operations (UK Patent ID: 6552820).

Robotics Lab, Indian Institute of Technology, Dhanbad — Dhanbad, India
Undergraduate Researcher under Prof Arun Dayal Udai — Sep 2023 – Feb 2025

- Developed a **6-SPS Parallel Robot for aseptic remote ultrasound** for AIIMS Jodhpur hospital in collaboration with Aduza.
- **Bachelor Thesis** on **6-DOF Stewart Gough Platform** for Potentiometer-Based Joystick using forward kinematic control
- Fabricated and controlled both the Stewart Platform using **Inverse/Forward kinematic modeling, CAE, Waterjet, 3D printing, Welding, circuit design, actuator motion control**. Deployed both the parallel robots for industrial use.
- Used **Yaskawa GP-12** industrial arm for increasing robot accuracy. **Patented Granted** for the Stewart Joystick (UK ID: 6552818).

M2D2 Lab, Indian Institute of Science (IISc) — Bengaluru, India
Research Intern under Prof. G. K. Ananthasuresh — May 2023 – Jul 2023

- Conducted Research on **Compliant Mechanisms** and **Kinematic modeling** on MedTech products assisting the elderly
- Assisted Translead medtech for developing 6-bar slider-crank mechanism for a responsive compliant chair – Stand-At Ease, assisting elderly in stand to sit motion. Did **Kinematic and dynamic motion analysis** via MATLAB and Msc Adams.
- Also developed and perfomed FEM analysis for a **compliant mechanism based grip strength analyser** (UK Patent ID 6552811).

INTERNATIONAL PATENTS (ALL GRANTED)

Integrated 6-bar Kinematic Harvesting Tool with Grappling Mechanism for Efficient Pedicle Cutting — Details
UK Patent ID: 6552819 ***Shobit Verma***, *Ping-Lang Yen*

Developed a single-DOF, 6-bar kinematic harvesting tool integrating synchronized grappling and cutting mechanisms.Derived governing mathematical model by resolving 40 parameters to predict the hook trajectory across three phases (hooking, pulling, holding) for autonomous harvesting.

Design and Development of 6-DOF Stewart Gough Platform for Potentiometer-Based Joystick — Details
UK Patent ID: 6552818 ***Shobit Verma***, *Arun Dayal Udai*

Developed a 6-DOF Stewart-Gough platform with forward kinematic control, integrating six potentiometer arms and specially designed 3D wedges to capture translational and rotational inputs, enabling real-time pose control of large parallel manipulators for robotics and aerospace applications.

A Robotic RCM-Based Needle Insertion System for Autonomous Abdominal Tumor Resection — Details
UK Patent ID: 6552820 *Ping-Lang Yen*, ***Shobit Verma***, *Yuting Lo*

Developed a single-DOF Remote Center of Motion (RCM) mechanism for surgical needle insertion, integrating parallelogram-based angular alignment, gear-operated locking, modular needle deployment, and manual override to enable controlled needle positioning during tumor resection procedures.

Monolithic Compliant Mechanism-Based Hand Grip Strength Analyzer — Details
UK Patent ID: 6552811 ***Shobit Verma***, *Sangeeth M. Ajith*

Developed an innovative monolithic compliant hand grip strength analyzer that accurately measures grip force through elastic deformation and indicator displacement, eliminating conventional joints and springs while offering adjustable stiffness, high repeatability with minimum error

Robotic Load-Haul-Dumper (LHD) Scaled System for Mining Automation Technology Development Details

UK Patent ID: 6552813 ***Shobit Verma***, *Bryan Vas, Bhupendra Singh Rathore*

Developed an autonomous, center-articulated Load-Haul-Dumper (LHD) prototype integrating modular chassis design, hydraulic steering, and ROS-based navigation with A* path planning and predictive control, offering a cost-effective, customizable platform for mining automation.

Retractable Manhole Slurry Agitator for Waste Reservoir Homogenization Details

UK Patent ID: 6552817 *Aaromal S. Ram, Bryan Vas, Rukhsar,* ***Shobit Verma***

Developed a retractable slurry agitator with a 1 - 5 m extendable telescopic shaft, automated blade engagement, PLC-controlled extension and rotation, and rotary encoder feedback, designed for being able to homogenize high-viscosity slurry in deep and narrow sewage reservoirs.

Domestic Patents

6-DOF Parallel Kinematic Robot for Respiratory Motion Simulation in Human Anthropomorphic Phantoms Details

Indian Patent ID: 519664-001 (Filed) ***Shobit Verma***, *Ping-Lang Yen, Zuo Haocheng*

Developed a 6-DOF, 6-UPS Stewart platform comprising of 340+ parts, six linear actuators, and custom joints and wedges, integrating inverse kinematics and machine-learning-based patient respiratory data to replicate breathing motion for robot-assisted surgical research and testing.

Self-Balancing Robotic Bike with an Omni-Directional Wheel Drive System Details

Indian Patent ID: 519666-001 (Filed) *Shailesh,* ***Shobit Verma***, *Aniket Bodele, Seloni Sinha, Sanskar Shukla*

Developed a self-balancing robotic bike with dual omni-directional wheels, deriving nonlinear dynamics using Lagrangian mechanics and implementing LQR feedback control for upright stabilization and multidirectional motion, validated through simulation and physical prototyping.

24-Bar Kinematic Mechanism with 1-DOF Dual-Phase Coupling for Gait Motion Synthesis Details

Indian Patent ID: 519665-001 (Filed) ***Shobit Verma***

Developed a single-DOF, 24-bar kinematic mechanism replicating cheetah-like running motion through mirrored linkages, revolute joints, and gear-driven phase synchronization, generating coordinated leg trajectories from a single rotary input to achieve quadrupedal locomotion.

Upcoming Research Papers

Electronic Borehole Compass for Real-Time Borehole Trajectory Measurement and Borehole Survey Details

In collaboration with Coal India Innovation Center *Santu Kumar, Rabindra Kumar Sinha,* ***Shobit Verma***, *Sikandar Kumar, Nishant Sinha, Sameer Sagar Sen*

Developed an electronic borehole compass integrating accelerometer, gyroscope, and magnetometer sensors with sensor-fusion algorithms, Bluetooth connectivity, and 3D trajectory reconstruction for real-time borehole surveying, field-tested in underground mines at depths exceeding 200 m.

Development of an Autonomous Robotic Surgical System for Minimally Invasive Tumor Surgery Using a Respiratory Motion-Controlled Anthropomorphic Phantom Details

Under RMML Lab, National Taiwan University *Ping-Lang Yen, Zuo Haocheng,* ***Shobit Verma***, *Cheng Yen Chung, Yuting Lo, Naved Akhtar, Giovanny Limin*

Developing an autonomous surgical system integrating robotic arm manipulation, precision needle insertion, and Stewart platform-based respiratory simulation, using intelligent control algorithms for motion-compensated tumor targeting and minimally invasive surgery on anatomical phantoms.

Awards

- **1st Rank** in Society of Automotive Engineers India's National EV Design Challenge among **279 teams**.
- **1st Rank** in drone building event by Ministry of Electronics and Information Technology (MeitY).
- **Bronze Medalist** in Inter IIT Tech Meet 12.0 for Solinas Product Design Challenge.
- **1st Rank** in EV Thon, a prototyping event in Concetto '22, The technical fest of IIT Dhanbad.
- **2nd Rank** amongst **103 teams** in CADastra, CAD design competition in Prometeo '23 IIT Jodhpur.
- **2nd Rank** in Designosis, Bike design challenge in Concetto, The technical fest of IIT Dhanbad.
- **3rd Rank** in Dezyn-o-mania, a machinery design competition in Khanan, geo-mining fest of India.
- **Exam scores:** Top 0.8% in JEE exam amongst **1 million+ candidates**, IELTS band 8.5 overall, TEEP AsiaPLus Scholar

Technical Skills

C, C++, Python, SolidWorks, Creo, Fusion 360, AutoCAD, ANSYS, MATLAB/Simscape, Blender, Mastercam, ROS, OpenCV, SAP, GEMS, CNC, VMC, 3D Printing, Pneumatic Control, GD&T, FEA, CFD, Kinematic and Dynamic Modelling, Failure Analysis.

Extra-curricular

- **Senate Member – Student's Gymkhana:** Won the annual college student governing body elections.
- **Team Lead – Design Ingenesis:** Founded the mechanical society of IIT Dhanbad.
- **Student Intern Representative:** Mechanical Engineering '25 batch.
- **Arts:** Pottery, Sculpting and Traditional Indian flute playing.